



Release date: Enquire March/April 1978



NewBear Computing Store

A Division of NEWBURY LABORATORIES LTD.



SALES & SERVICE: BONE LANE NEWBURY BERKSHIRE RG14 5SH Tel: 0635 46898 Telex: 858815

PRICE LIST

77-68 MON. 1 BOARD

(Bootstrap loader for "Soft" Monitor)

1 off	7406	0.64	2 off	81LS97	2.86
3 off	74LS00	0.63	1 off	ROM 1	3.00
3 off	74LS04	0.78	(1 off	ROM 2	3.00 optional)
1 off	74LS10	0.21	1 off	6850	8.83
1 off	74LS14	1.26	(1 off	6850	8.83 optional)
1 off	74LS20	0.21	8 off	2102-1	12.00
1 off	74LS90	0.92	2 off	2N2907A	0.74
1 off	74LS93	0.92	2 off	BC107	0.44
1 off	74LS139	1.05	2 off	1N4148	0.12
2 off	74LS161	2.44	2 off	36v ZENER	0.50

37.55

VAT @ 8% 3.00

4 off	47)	0.16
2 off	100) $\frac{1}{2}$	0.08
2 off	22) watt	0.08
2 off	470)	0.08
2 off	820 1 watt	0.10
15 off	4.7K) $\frac{1}{2}$	0.60
2 off	10K) watt	0.08
		1.18
VAT @ 12 $\frac{1}{2}$ %		0.15

CAPACITORS

2 off	0.001uF)	0.16
4 off	0.01uF)	0.32
2 off	0.047uF)	0.16
7 off	0.1uF)	0.77
3 off	10uF taut bead	0.60
2 off	47uF electro-lytic.	0.22
		2.23
VAT @ 12 $\frac{1}{2}$ %		0.28

SOCKETS & MISCELLANEOUS

2 off	16 pin D-I-L	0.50
(8 off	16 pin D-I-L	2.00)
1 off	24 pin D-I-L	0.36
(1 off	24 pin D-I-L	0.36)
1 off	78 way edge connector	1.80
1 off	p.c.b.	10.00
3'	Ribbon Cable (solid core)	0.60
1 off	single pole switch	0.69
		14.55
VAT @ 8%		1.11

TOTAL INCLUDING £4.56 VAT & 30p P&P £60.38

ITEMS IN BRACKETS ARE OPTIONAL AND NOT INCLUDED IN OVERALL PRICES.



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MICROCOMPUTER DESIGNS

77 - 68 The Construction of a simple Microcomputer using a 6800 Microprocessor

- * Needs no expensive extras such as a Visual Display Unit etc.
- * Uses east to obtain TTL logic plus one cpu I C and two memory I C's.
- * Designed for easy construction.
- * Can be expanded.
- * User group support.
- * Spare sets of diagrams are available.
- * Printed Circuit Boards are available.

Description

This microcomputer is the minimum cost method of obtaining a working 6800 based system. The 78 way connector allows full expansion from its initial 256 bytes of memory to the full 65 Kilo Bytes. The equipment practice is based on the 8 inch by 8 inch p.c.b. and 0.1" gold plated edge connectors (which can be easily obtained in Europe). To start only one board is required. The component count is as follows:-

19	TTL 74 series I C's	9	LED's
2	Buffers (optional) I C's	17	Toggle Switches
2	Memory I C's	2	Push Button Switches
1	6800 cpu I C	1	p.c.b.
48	Resistors	1	Power Supply + 5V
		1	Box

A monitoring ROM has not been included in order to keep the cost down and to give the user complete freedom of choice for his monitoring system. Full details of how to use this type of system are included and the advantages and disadvantages debated.

To help the constructor with moral support, news of additional designs and p.c.b's that become available, a User group has been formed. This will issue a quarterly newsletter containing a register of users names and addresses, and an Index of additional Design Notes and software. The first year's membership will be free with the purchase of the book.

Most constructional diagrams have a hard life - They become very dog-eared and covered in additional marks (pencil, coffee, solderflux etc.) To save your book from these natural hazards spare sets of diagrams can be purchased.

Any comments on the text or points of clarification that might assist others should be sent to the User group. In this way others may benefit from your experiences.

STOP PRESS

- * Bootstrap Loader Board available (Mon 1)
- * 4K byte RAM Board available



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PRICE LIST

77-68 COMPONENTS

LOGIC

4 off	7400	0.64	1 off	7495	0.65
1 off	7402	0.16	2 off	74125	1.35
1 off	7404	0.24	2 off	74157	1.98
1 off	7410	0.18	2 off	81LS97	2.86
2 off	7428	0.86	2 off	2112	6.08
1 off	7430	0.18	1 off	6800	14.00
2 off	7432	0.64			
1 off	7474	0.34			
2 off	74LS75	1.10			
			TOTAL		31.08

RESISTORS & CAPACITORS VAT 12½%

20 off	4.7K	0.80	3 off	47uF	0.33
2 off	1K	0.08	8 off	0.1uF	0.88
14 off	430	0.56	1 off	33pF	0.05
1 off	270	0.04			
2 off	220	0.08			
2 off	22	0.08			
			TOTAL		2.90

LED's VAT 12½%

9 off 1.35p

LOW PROFILE SOCKETS

2 off 16 pin 0.50p
1 off 40 pin 0.66p

BOARD

77-68 p.c.b. 10.00
80 way edge connector 1.80

BOXES 14½" x 13½" x 3"

a) To contain single card and the power supply £3.00
Drilled & engraved front panel 14½" x 2½" x 1/8" Al £7.00
Drilled back panel 14½" x 2½" x 1/8" Al £3.00
4 chromed handles to fit above £5.00
POSTAGE & PACKING £1.50
£19.50

OR

b) 19" 5u rack to allow for full expansion of 77-68
Vero type 3A No. 33-7111D-72666H £23.40
POSTAGE & PACKING £.60
TOTAL 24.00

VAT @ 8% unless *=VAT @ 12½%

Cheque to be made out to NEWBEAR COMPUTING STORE

Overseas orders please add an extra 50p for additional postage.

Postage and Packing 30p unless otherwise stated.



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PRICE LIST

77-68 4K RAM BOARD

LOGIC

2 off	81LS97	2.86		
1 off	74LS20	.21		
1 off	74LS86	.38		
1 off	74LS139	1.05		
32 off	2102-1	48.00	(or 2102-1 £51.20)	
	VAT @ 8%	4.24		£57.26

RESISTORS & CAPACITORS

1 off	1K ½ watt	0.04		
2 off	47uF 10v electrolytic	.22		
3 off	0.1uF ceramic	.33		
8 off	10uF 16v taut	1.60		
8 off	0.01uF ceramic	.64		
		2.83		
	VAT @ 12½%	.35		£ 3.18

SOCKETS

32 off	16 pin D-I-L (optional)	8.00		
1 off	78-way edge connector	1.80		
1 off	printed circuit board	10.00		
1½'	ribbon cable	.30		
		20.10		
	VAT @ 8%	1.61		£21.61
1 off	Design Note 15 Circuit Diagram	.60		£ .60

TOTAL INCLUDING £6.22 VAT & 30p P&P

£83.07

EXTRA IF REQUIRED

2 off	Approx. 10"x4" Vero board 01-0015E (for back plane)	2.94		
	VAT @ 8%	.24		£3.18
1 off	8"x8" prototyping p.c.b. for 77-68 system	9.95		
	Vero 10-0155F			
	VAT @ 8%	.80		£10.75
1 off	8"x8" prototyping p.c.b. (high pack density)	10.91		
	VAT @ 8%	.87		£11.78



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Low Power Schottky TTL Price List.

74LS00	0.21	74LS132	.90
74LS01	.21	74LS136	.38
74LS02	.21	74LS138	1.05
74LS03	.21	74LS139	1.05
74LS04	.26	74LS151	.96
74LS05	.26	74LS153	.96
74LS08	.21	74LS154	1.98
74LS09	.21	74LS155	1.05
74LS10	.21	74LS156	1.05
74LS11	.26	74LS157	.96
74LS12	.21	74LS158	.88
74LS13	.55	74LS160	1.22
74LS14	1.26	74LS161	1.22
74LS15	.21	74LS162	1.22
74LS20	.21	74LS163	1.22
74LS21	.26	74LS164	1.18
74LS22	.21	74LS168	2.03
74LS26	.31	74LS169	2.03
74LS27	.21	74LS170	2.61
74LS28	.40	74LS173	2.28
74LS30	.29	74LS174	1.18
74LS32	.24	74LS175	1.10
74LS33	.40	74LS189	2.82
74LS37	.30	74LS190	2.02
74LS38	.30	74LS191	2.02
74LS40	.27	74LS193	1.86
74LS42	.88	74LS196	1.21
74LS47	.96	74LS197	1.21
74LS48	.96	74LS247	.96
74LS51	.21	74LS248	.96
74LS54	.21	74LS249	.96
74LS55	.21	74LS253	1.05
74LS73	.34	74LS257	1.09
74LS74	.38	74LS258	1.09
74LS75	.55	74LS266	.38
74LS76	.34	74LS283	.96
74LS78	.34	74LS289	1.05
74LS83	1.05	74LS352	.96
74LS86	.38	74LS353	.51
74LS107	.38	74LS365	.51
74LS109	.38	74LS366	.51
74LS112	.38	74LS367	.51
74LS113	.38	74LS368	.51
74LS114	.38	74LS386	.38
74LS125	.56	74LS670	3.04
74LS126	.56		

All IC's are 100% guaranteed and manufactured by Texas, Signetic, National etc.

V.A.T. at 8%

30p p & p U.K. orders

50p p & p Overseas orders



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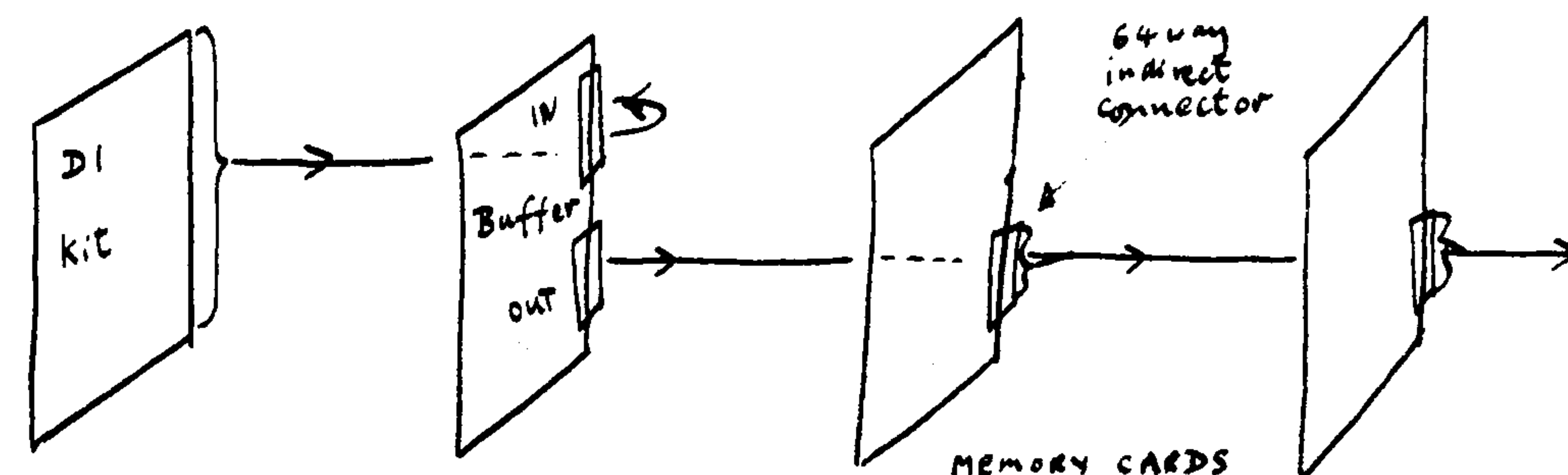
Memory extension for the MEK 6800D1 kit.

Introduction

The D1 kit unfortunately has an unbuffered bus which limits the number of cards that it can drive reliably to three. If the memory cards come in 4K blocks this means a limit of 12K bytes.

This problem can be overcome by fully buffering the appropriate part of the bus by changing standards and adopting the double Euro card. This has the advantages of using readily available connectors, prototyping cards, memory p.c.b.'s & racking systems. The circuit diagram for the buffering is Design Note 16 and for the memory board, Design Note 11.

The practical layout of a fully buffered system is:-



Double Eurocard solution

This memory card was designed to be a general purpose 4K RAM card and to be available at an economic price. To achieve this end the address decoding etc. has to be hand wired on a prototyping area. See the 1/9/77 catalogue under Design Notes 11, 12, & 13 for full description. Also the board does not have plated through holes in the memory block area. This means the memory IC's need soldering both sides of the board. If sockets are required, the "soldercon" type must be used.

Exorciser Compatible 4K RAM card solution

If for various reasons the change of card size & connector type is unacceptable, there is an alternative. This card conforms with the Exorciser & D1 kit bus & is a printed circuit for both the memory & address decoding etc. blocks. Again to keep the cost to a minimum plated through holes have not been used & Z links etc. must be wired in.

For fully assembled & tested boards contact the "Bear".



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4K Ram board. Exerciser compatible using .156" connectors

Logic

2 off	8T26	3.68
2 off	8T 97	3.20
2 off	74LS138	2.10
1 off	74LS02	.21
32 off	2102-1	48.00
	(or 2102L-1)	(51.20)

VAT @ 8%
57.19
4.58

61.77

Resistors & Capacitors

1 off	1k $\frac{1}{4}$ w	0.04
1 off	100uF electrolytic	0.12
14 off	0.1uF bead Tantalum	2.80
		2.96
	VAT @ 12%	37

3.33

1 off	p.c.b. ref. 4K H.D. 76	10.00
	VAT @ 8%	80
1 off	Design Note 14.	0.60
	Circuit Diagram & Constructional Details	

Total including 30p p & p & £5.95 VAT

0.156 edge connector £4.20

£76.82



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4K byte Static Memory Board for the MEK 6800D1 Evaluation kit.

Double Eurocard.

4 off	74LS04	£ 1.04
1 off	74LS32	.24
1 off	74LS10	.21
2 off	81LS97	£ 2.86
1 off	74LS85	£ 1.55
1 off	74LS155	£ 1.05
32	2102-1	£48.00

8	4.7K resistors	.36
8	switches (D-I-L)	£ 2.12
10	Decoupling Capacitors	

	.1uF	£ 1.10
1 off	Euro plug	£ 1.65
1 off	Euro socket	£ 2.99
1 off	p.c.b.	£ 6.25
	Connecting Wire	.50
1	diagram	.60

£70.52

p & p 30

VAT 8% £ 5.67

£76.49



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MICROCOMPUTER DESIGNS

WB-1 - The Design and Construction of a simple Computer

- * Needs no expensive extras such as Visual Display Units etc.
- * Uses easy to obtain TTL logic in a proven design.
- * Designed for home construction.
- * A detailed introduction to the hardware of 8 bit microcomputers.
- * User group support.
- * Spare sets of Diagrams are available.

Description

The major difficulty in using small computers is to understand how they work. To the newcomer they are horrifically complex. After the first few sentences the jargon becomes incomprehensible.

A short lecture course may explain many mysteries but can never be as enlightening as first-hand experience. WB-1 is not to be confused with a computer using a microprocessor. It is designed at a more fundamental level using TTL logic. The advantages are that the level of understanding reached is more profound and that modifications can easily be made even to the instruction set. This is not the case with a microprocessor. The disadvantages are that there is more wiring involved in connecting the logic together. The cost is heavily dependant on access to a good junkbox. It can vary from nothing to around £50.

To keep the cost down the memory is limited to 256 bytes and the controls are toggle switches and L.E.D. lamps. These controls can be expanded to the limit of the constructor's imagination and finance.

To help the constructor with moral support, additional hardware designs for expansion and software, a usergroup has been formed. At quarterly intervals a register of members and index of articles will be circulated. Any corrections, or clarifications to the book will appear in the user group literature. The first year's membership of this group is free with the purchase of the book.

Most constructional diagrams have a hard life - they get covered in comments, ticks, coffee stains and soldering marks. To overcome these natural hazards, spare sets of diagrams can be purchased.

Any comments on the text or points of clarification that might assist others should be sent to the user group. In this way others may benefit from your experiences.

Book reference DC1
Spare Diagrams reference DC1A

(90 pages A4)

1/5/77



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Systems Section

Provisional Specification

Cub.

- * 6800 based Microcomputer
- * touch keyboard
- * 64ch x 16 line CRT display
- * Kansas City 300 baud cassette interface
- * 4K Memory
- * Portable, "student proof" case
- * Bus assessable for memory extension
- * Bug bear Monitoring ROM (Mikbug compatible)
- * Fully assembled and guaranteed for 1 year
- * Provisionally costed at £480.00.

Panda

- * 6800 based microcomputer
- * Reed switch keyboard for "professional" use.
- * 64ch x 16 line CRT display
- * Kansas city 300 baud cassette interface
- * V24/RS232 C, interface (20mA correct loop adapter available)
- * 4K Memory
- * Portable, "student proof" case
- * Bus available
- * Optional 8K ROM & RAM boards available.
- * Bug bear Monitoring ROM Mikbug compatible with:
 - "B" function to access ROM e.g. Basic etc.
 - "T" function to act as a VDU through the V24 port. (e.g. teletype replacement).
 - "C" function to disable keyboard/CRT and behave as a microcomputer through the V24 port.
- * Fully assembled and guaranteed for 1 year
- * Provisional cost £630.00.

Availability of Cub: Enquire March 1978

Availability of Panda: Enquire April/May 1978

The "Bear" reserves the right to change this specification without notice.